Segmentation of SEM Fractograph into Void and Surface Regions

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Abstract. In a material, new surfaces are evolved and voids (holes) are formed due to fracture. To study the characteristic of a fracture surface, the analysis of the void distribution is very important. The inception of Scanning Electron Microscope (SEM) has motivated the researchers towards the quantitative analysis of such surfaces. Segmentation of SEM fractograph into surface and void regions is the fundamental step to analyze the void distribution. So far, the image processing tools are being applied by the researchers mostly on a case to case basis and the success depends on the understanding of the end user about the basic image processing operations offered by the tool. Thus, a well founded composite tool to cater the specific need is still lacking. In this work, we have proposed a digital image processing based automated scheme to extract surface and void regions.

Key words: Fractograph Analysis, Fractograph Segmentation, Surface Boundary Detection, Void Analysis, Void Detection

1 Introduction

In material research, quantitative fractography is an analytical tool for estimating the feature characteristics and topography of fracture surfaces. Early meaningful applications on steel [1, 2] showed that new surfaces originating from fracture at varying conditions differ significantly in their quantitative geometric characteristics. But such differences are indistinguishable in qualitative inspection.

Fracture of load bearing components while in service occurs for various reasons. The new surfaces evolved due to fracture carry the impression about the mode of fracture. It also manifests the micro-structure present in the material and the loading conditions the material has experienced. Optical microscope has limited capability in capturing the details required to carry out the analysis. The inception of Scanning Electron Microscope (SEM) has enabled the researcher to go for quantitative measurements on SEM fractograph.

The common way to characterize a fracture surface is by quantifying the percentage area of the fracture surfaces covered by ductile and brittle fracture characteristics. The steps involved in the ductile fracture are void nucleation, void growth and void coalescence. Initiation and growth of voids are caused by inclusions, precipitates, and other second-phase particulate matter under the influence of plastic strains and hydrostatic stress [3–6]. The role of inclusions and particles in initiating ductile fracture has been demonstrated for a variety of materials like oxidized copper alloys, maraging steel, low-strength steels, aluminum alloys. The distribution, size, shape, type, and coherency of these constituents of the micro-structure play an important role in controlling void formation and eventual fracture [7–10]. For ductile materials, the engineering properties are determined by the interaction of stress and strain fields with the micro-structure of a material. The contribution of deformation processes in the development of voids has been well established. While the growth mechanisms can vary, say, with the temperature of testing [11] their nucleation invariably occurs where inhomogeneous deformation takes place.

Various attempts have so far been made to correlate the fracture characteristics with micro-structure. It has been indicated that the size, shape and distribution of the voids on fracture surfaces are related with the size and distribution of second phase particles from where the voids are nucleated. They also depend on loading rate and temperature [12–14]. Das et al. [15, 16] have showed good correlation between void morphologies with tensile properties of copper strengthened HSLA steel and 304 LN austenitic stainless steel.

Past study indicates that proper analysis of fractographic features can yield valuable information for correlations of micro-structural parameters with the micro-mechanisms of the fracture processes. In order to develop an automated tool, there is enormous scope to apply digital image processing technique on Fractograph. As the first step, the image has to segmented into surface and void regions to carry out the subsequent study of their characteristics and finally the quantification of the properties. Image processing techniques are also being deployed by the researchers [16] but the efforts are limited to the manual application of the general image processing tools on adhoc manner and the outcome depends on the knowledge of the individual. It has motivated us to develop an automated process to serve the purpose. The paper is organized as follows. After this brief introduction, proposed methodology is presented in section 2. Experimental results are shown in section 3. Finally, the concluding remarks are put into section 4.

2 Proposed Methodology

A fractograph image shows various surfaces and the voids present. The voids may vary in size. Physically, a significant void has a considerable depth with minimum intensity at the bottom and ideally, the intensity gradually increases till it meets the surface. On the other hand, a small void may be indicated by only a small patch of dark intensity without any noticeable depth. A fractograph image also shows the boundaries which are formed when two different surfaces meet each other or a significant void meets the surface. Boundaries posses maximum intensity with white appearance.

In our effort, we first try to extract boundaries to form the closed contour of the regions. Subsequently, the enclosed regions are categorized as void and surface. It may be noted that a surface may have one or more small voids. The proposed segmentation scheme consists of number of stages as follows.

- Detection of region contour
- Refinement of region contour
- Region Characterization

The complexity in the surface undulation, diversity in their absorption and reflective property, wide variation in void depth and pattern lead to considerable deviation between the ideal scenario for different components (void, surface, boundary) and the actual one reflected in the fractograph images. Thus, the task becomes non trivial. The details of the various steps are presented in the following sub-sections.

2.1 Detection of Region Contour

Ideally, boundary pixels will have maximum intensity value and there exists a noticeably sharp transition in intensity value. It seems a simple threshold or edge detection technique may help us in determining the boundaries. But actual images reflect that the pixels of a surface/void around the boundary may also possess intensity values closer to those in boundary. Furthermore, the pixels in the surface/void may have considerable intensity variation. As a result a simple threshold or conventional edge detection techniques directly can not provide the desired output. Plenty of additional edges may come up. Moreover, the boundary will not be a closed one as all the pixels along the boundary may not have the desired high intensity values. Thus, detection of closed contour is quite challenging and we have relied on the technique presented in [17].

A series of steps are followed to extract the contour. The gray-scale fractograph image is first binarized following intensity based thresholding. The threshold value is chosen by analyzing the intensity histogram for the image (for details refer [17]). The pixels with intensity value higher than the chosen threshold are taken as the boundary pixels.

The boundaries thus obtained after thresholding may have discontinuities and spurious pixels. Further processing is required to obtain the closed region

contour. As the boundary segments may be thick, a thinning algorithm is carried out to make it single pixel width. Edge linking is applied on the thinned image so that broken edges are glued. A Pixel is linked with another in a predefined neighbourhood having closest gradient direction. In case of non-availability of any pixel in the said neighbourhood, then its size is increased up to a maximum.

After linking, surface/void contours obtained are mostly closed unless the boundary obtained after thresholding had a wide discontinuity. We may still have curve segments which at this stage are unlikely to form a closed contour. Such open curves are pruned out following a region growing technique.

The closed contour thus obtained relies heavily on the global feature of the image. The boundary pixels are obtained after thresholding and threshold is selected based on the intensity histogram of the entire image. Due to the variation present in the image, an actual boundary segment may be lost if it fails to clear the threshold. May be, some of the pruned segments also could not form the closed contours because of the segments lost during threshold operation. It has forced us to further refine the contour. Fig. 1 shows a fractograh image and the extracted contour. One of the areas that requires contour refinement has been marked in it.

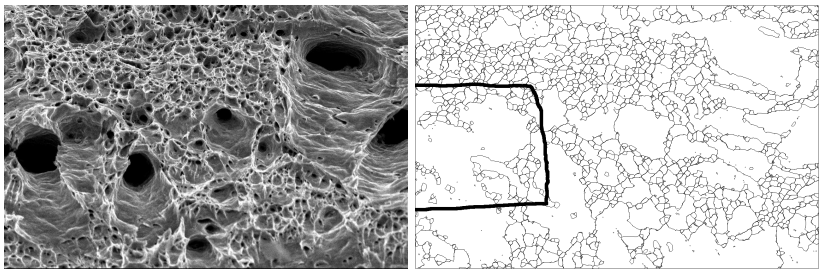


Fig. 1. A fractograph image and extracted region contour with one marked area that needs refinement

2.2 Refinement of Region Contour

Detection of region contour as discussed in section 2.1, ignores the local intensity variation in and around the regions. As a result, weak boundary segments have been missed out. Thus, it becomes essential to refine the contours. At this stage, we can concentrate on each individual contour and thereby, it becomes easier to take the local intensity information into consideration.

It has been observed that in a fractograph image, most of the regions are of uniform size with little bit variation. Regions with void(s) of considerable depth are comparatively bigger in size. As the deformation leads to formation of voids, their growth and coalescence of those thereafter gives rise to larger regions. Moreover miss in detecting boundaries at first stage also may give rise

to larger regions. With these observations, we carry out the contour refinement process only in the large regions.

The candidate regions that may undergo further splitting are chosen as follows. Let μ_s and σ_s are the average and standard deviation of size of the extracted regions. Let S_i be the size of the i-th region. It is considered to be a candidate if $S_i > \mu_s + \sigma_s$. For each such candidate region, the steps described in section 2.1 are repeated to explore the possibility of further decomposition. At this stage, threshold is chosen by considering the local intensity distribution of the region. As it has been discussed, ideally a large region is caused by a void of considerable depth. Pixels in a void are having low intensity value with minimum at bottom and gradually higher value as it approaches the surface. The boundary pixels will have comparatively higher value. But, they may not be always high enough as the chosen threshold value in first stage. To overcome this, we consider $\mu_l + .5\sigma_l$ as the threshold value, where μ_l and σ_l denote the average and standard deviation of the intensity value in the candidate region respectively. Pixels with intensity values higher than the threshold are taken as the boundary pixels. Thinning, linking and pruning(region growing) steps as in section 2.1 are then carried out with the newly detected boundary pixels to find out the final contour. It may noted that threshold value chosen is quite liberal to exclude the possibility of missing any likely boundary pixels. On the other hand, during linking, neighbourhood size is reduced to restrict the over splitting of the region. Spurious boundary pixels will fail to form a closed contour and thereby will be pruned out subsequently. In Fig. 2, the refined contour corresponding to the image shown in Fig. 1 has been presented. The marked area shows the desired decompositions which were missed earlier.

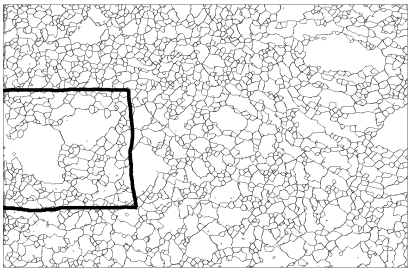


Fig. 2. Refined region contour corresponding to the image shown in Fig. 1

2.3 Region Characterization

With the refined contour of the regions, the task of characterizing the regions is carried out. Broadly, the regions are characterized either as void or as surface. It has been observed that another type of regions also exists which comprises of surface and one or more voids with negligible depth. Thus, to be more specific,

regions are categorized into three classes namely, *void region*, *surface* and *mixed region*. A void region consists of a void of considerable depth (type-1 void). A surface region has no void. On the other hand, a mixed region comprises of surface and void(s) of insignificant depth (type-2 void).

The major challenge is to detect the void regions. Such regions can ideally be characterized by the presence of intensity gradient along its depth. But, similar gradient may also be observed within a surface. As a result, we have relied upon different signature to discriminate the classes of regions.

The task of region characterization starts with void detection. It consists of two steps: detection of void tips and its extension. It has been observed that both the types of void have a zone with very low intensity value. We refer to such zone as the tip *i.e.* the bottom most area of a void. The intensity value, I_t corresponding to the local peak in the low range of the intensity histogram of the image can be used in determining the tip of the voids. The intensity histogram is formed by dividing the intensity scale into bins of small size and local maxima is determined by considering the previous and next bin. The upper bound of the intensity bin corresponding to the first local maxima is taken as I_t. Pixels with intensity value smaller than I_t form the tip.

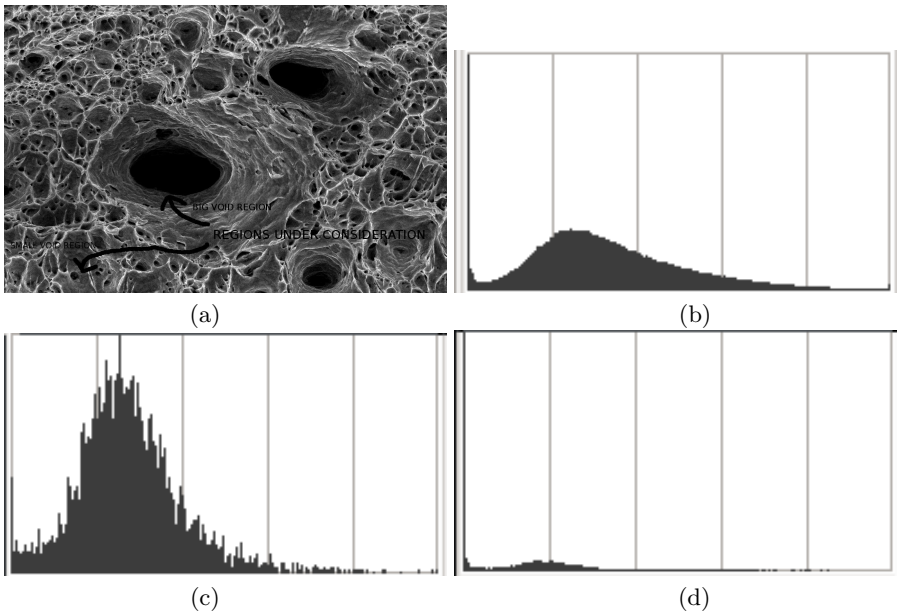


Fig. 3. (a) A fractograph image with two marked voids (with and without considerable depth), (b) Intensity histogram corresponding to the image, (c) Intensity histogram corresponding to marked void without considerable depth and (d) Same for marked void with considerable depth

In Fig. 3, a fractograph image has been shown. Two voids have been marked of which one belongs to type-1 and the other one belongs to type-2 void. Intensity histograms of the image and the marked voids have also been shown. From the histograms, it is clear that for type-1 void *i.e.* void of significant depth, quite a large number of pixels form the tip and it is not so for type-2 void. In order to include those missing pixels of the void, extension of the void tip is essential.

The second step of void detection *i.e.* extension of void tip is performed only on the regions with void tip. A void, apart from the tip, also consists of other pixels surrounding the tip. Physically, it may be thought of as the extension of void or the void pixels at higher levels along its depth. Such pixels have higher intensity value but should not be confused with the surface pixels. This is very crucial as the scenario is different for a void and a mixed region. The intensity of such pixels may also vary across the void depending on its depth and neighbourhood in case of a mixed region. Hence, a global threshold can not cope up with the situation. To deal with this sensitive task, again we rely on the local intensity pattern of the region. A threshold, I_v is taken as $max(\mu_l - .5\sigma_l,\ I_t)$. μ_l and σ_l are average and standard deviation of the intensity in the region respectively. Pixels with intensity value smaller than I_v are taken as the part of the void.

As it appears in the histograms shown in Fig. 3, for type-1 voids unless a substantial number of pixels are lying at lower depth (towards the surface level), μ_l is very close to I_t. As a result, effectively I_v will be either same as I_t or a value closely around it. Thus, its scope for further extension is limited. Mostly, such type of voids are detected at first stage. On the other hand, μ_l for type-2 voids will be higher than I_t and I_v may take the value of $\mu_l - .5\sigma_l$. Thereby, a relaxed threshold will enable the extension of the tips.

Once the voids have been detected, the classification of the regions is done. The regions without any void can be readily identified as the surface. A region fully covered by a void can be treated as a void region. But, it may not be true for all void regions. Hence, for the regions partially covered by the void have to be analyzed further for their actual classification as mixed or void region. Let I_p be the intensity corresponding to the peak in the intensity histogram of the region. I_t, as discussed earlier, stands for the intensity corresponding to the first local maxima in the intensity histogram of the image. For a void region, $I_p <= I_t$ and it is unlikely to be true for a mixed region because of the presence of surface pixels of higher intensities. In order to reduce the possibility of misdetection of a mixed region, further analysis is made. Let I_c be the minimum cut-off intensity value such that $k\%$ pixels in the region have the intensity lower than the cut-off and μ_g, σ_g are the average and standard deviation of intensity values for the whole image respectively. In order to qualify for a void region, I_c must be smaller than $\mu_g + j\sigma_g$ which implies that in the context of the image $k\%$ pixels in the region are less likely to bear the signature of the surface. In our experiment, k and j have been taken as 80 and 1 respectively.

3 Experimental Results

A Titanium-stabilized Interstitial Free (IF) steel in the form of cold rolled sheets of 2.3 mm thickness received from TATA Steel Ltd., Jamshedpur, India, has been used in the present investigation. Two types of tests, tensile test and ratcheting test were performed on specimens made from the as received steel sheets. Strain controlled tensile test at a nominal strain rate of 10^{-3} sec^{-1} was done at room temperature (28^0C) under normal laboratory atmosphere in a computer controlled close-loop servo-hydraulic universal testing machine of $\pm 100kN$ load capacity, Instron 8501. Ratcheting tests were done by asymmetric cyclic stressing with different combinations of mean stress and stress amplitude at a stress rate of $250MPa$ sec^{-1} for different constant maximum stresses.

The specimens used for examining the features on fracture surfaces were obtained by sectioning the tensile fractured specimens (with and without prior ratcheting) on planes perpendicular to the tensile axis leaving a distance of $10mm$ from the fracture surfaces. Before inserting the sectioned specimens in the vacuum chamber of SEM the fracture surfaces were ultrasonically cleaned and observed in a Jeol make, Japan, Scanning Electron Microscope (JSM6360) under secondary electron imaging mode with an accelerating voltage of $20kV$ and under condition of zero tilt. The brightness and contrast of the images were controlled by the in-built image controlling software of the microscope. However, manual intervention was also done in controlling the brightness and contrast, wherever needed. Secondary electron images of the fracture surfaces from each of the sectioned specimens taken at three different magnifications ($\times$500, $\times$1000 and $\times$2000) were digitally recorded in a computer interfaced with the microscope. In total 45 secondary electron images were taken. The fractograph images thus obtained have been used in this work.

Sample results have been shown in Fig. 4. It has been observed that proposed methodology is successful in refining the contour of the regions extracted by following the techniques presented in [17]. Refinement has reinstated the missing boundaries that has enabled the classification of the regions. In Fig. 4, a white region stands for surface, contour filled with black and gray patch signifies a void region and for the mixed region contour is filled by white and black patch.

4 Conclusion

In this work we have presented an automated scheme for segmenting SEM fractograph images. An existing technique for extracting the region contour has been modified to take the local intensity pattern into consideration. As a result refined contour of the regions have been obtained and those provide better correspondence with the image. Based on the observation and subsequent analysis, voids are detected. Finally, regions are classified as surface, void and mixed regions. Experimental result indicates that the proposed methodology works well for images of considerable variety. In future, quantititave analysis of the void distribution may be carried out.

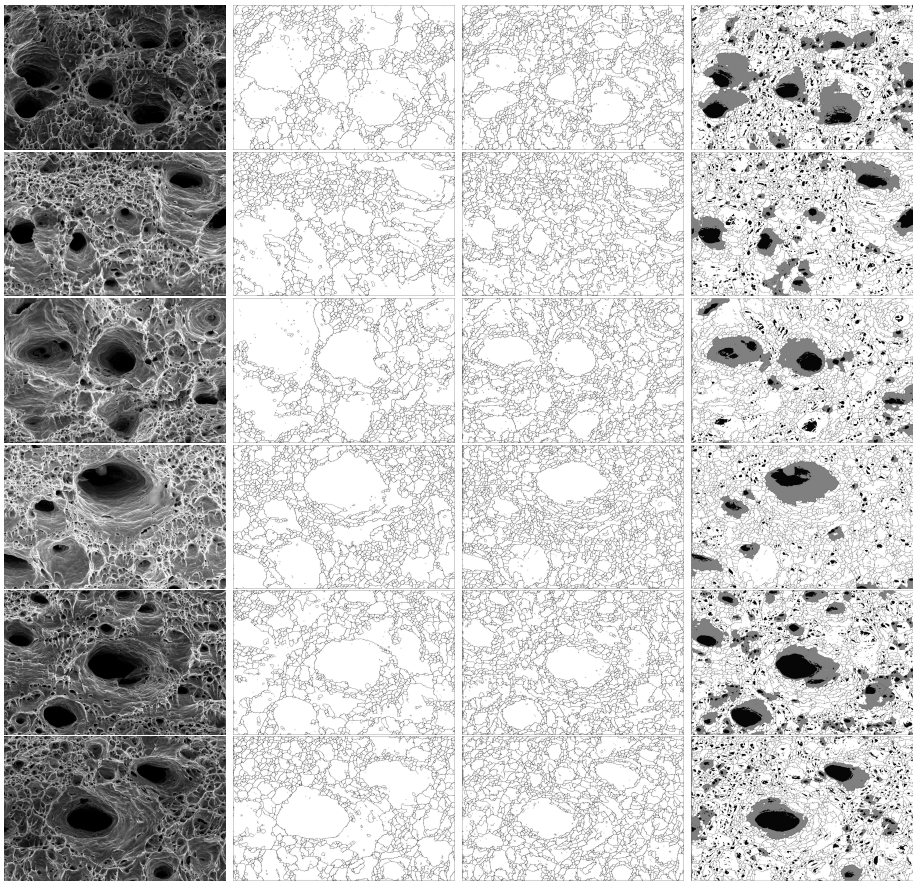


Fig. 4. Sample results:(a) Fractograph image (b) Extracted region contour (c) Contour after refinement (d) Classified region (white denotes surface region, black and gray filled area denotes void region, white area with only black patch denotes mixed region).

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